identify the global optimum, make an observation at the recommended location, add the new point to our training data, and repeat the entire procedure until we reach some condition for terminating.

SIDEBAR **Connection to Design of Experiments**

The description of Bayesian optimization at this point might remind you of the concept of Design of Experiments (DoE) in statistics, which sets out to solve the same problem of optimizing an objective function by tweaking the controllable settings. Many connections exist between these two techniques, but Bayesian optimization can be seen as a more general approach to DoE that is powered by the machine learning model Gaussian process.

Things might have gotten a bit confusing up to this point, so it is time for us to take a step back and look at the big picture of Bayesian optimization.

1.2.3 Combining the Gaussian process and the optimization policy to form the optimization loop

In this subsection, we tie everything that we have described so far and make the procedure more concrete. We will see the Bayesian optimization workflow as a whole and better understand how the various components work with each other.

First, we start out with some initial data set like those in Table 1.1, Table 1.3, and Table 1.2. Then the Bayesian optimization workflow is visualized in Figure 1.6, which is summarized as follows.

1. We train a GP model on this set, which gives us a belief about what our objective looks like everywhere, based on what we have observed from the training data. This belief is represented by the solid curve and shaded region like those in Figure 1.4 and Figure 1.5.
2. A Bayesian optimization policy then takes in this GP and scores each point in the domain in terms of how valuable the point is in helping us locate the global optimum. This is indicated by the green curve like in Figure 1.5.
3. The point that maximizes this score is the point that we will choose to evaluate the objective at next and is then added to our training data set.
4. The process repeats until we cannot afford to evaluate the objective anymore.

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